

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.4292631924734626, median 0.3086363687460063, std: 0.4013596632105671
Gyroscope error (imu0): mean 0.42136288598396043, median 0.36569957124873564, std: 0.2657215435807544
Accelerometer error (imu0): mean 0.4781474564836466, median 0.42272245761483335, std: 0.30230686015888086

Residuals

Reprojection error (cam0) [px]: mean 0.4292631924734626, median 0.3086363687460063, std: 0.4013596632105671
Gyroscope error (imu0) [rad/s]: mean 0.015258909282802673, median 0.013243161104266557, std: 0.009622634225405819
Accelerometer error (imu0) [m/s²]: mean 0.07306467845038365, median 0.0645953042739653, std: 0.04619485732140284

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[ 0.99982652  0.01300494  0.01333418  0.01790775]
 [-0.01316443  0.99984201  0.01194364 -0.00073518]
 [-0.01317675 -0.01211711  0.99983976 -0.00070591]
 [ 0.          0.          0.          1.          ]]
```

T_ic: (cam0 to imu0):

```
[[ 0.99982652 -0.01316443 -0.01317675 -0.01792362]
 [ 0.01300494  0.99984201 -0.01211711  0.00049362]
 [ 0.01333418  0.01194364  0.99983976  0.0004758 ]
 [ 0.          0.          0.          1.          ]]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)

0.008816270957429255

Gravity vector in target coords: [m/s²]

[-0.02318056 0.0040996 -9.80652175]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [599.0144910569156, 599.5793593987496]

Principal point: [330.4358247492389, 244.51650425749176]

Distortion model: radtan

Distortion coefficients: [0.10660040555500379, -0.19725984022276483, -0.0017422990584042355, 0.0022020513377890858]

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.08 [m]

Spacing 0.024 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 200

Accelerometer:

Noise density: 0.010805145755125

Noise density (discrete): 0.1528078367031585

Random walk: 0.0003854976741129608

Gyroscope:

Noise density: 0.0025606617446109967

Noise density (discrete): 0.03621322567878822

Random walk: 1.507779944779771e-05

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

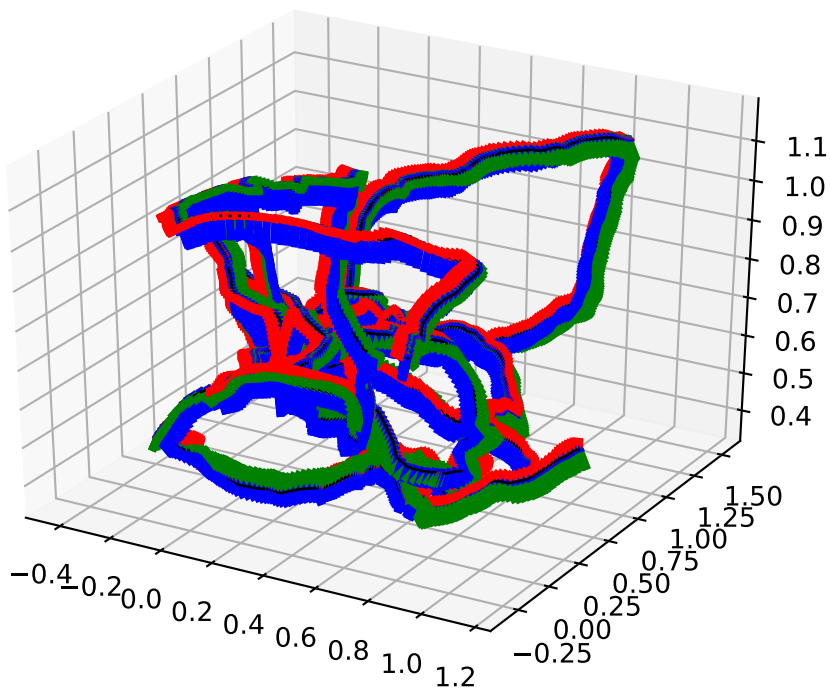
[0. 1. 0. 0.]

[0. 0. 1. 0.]

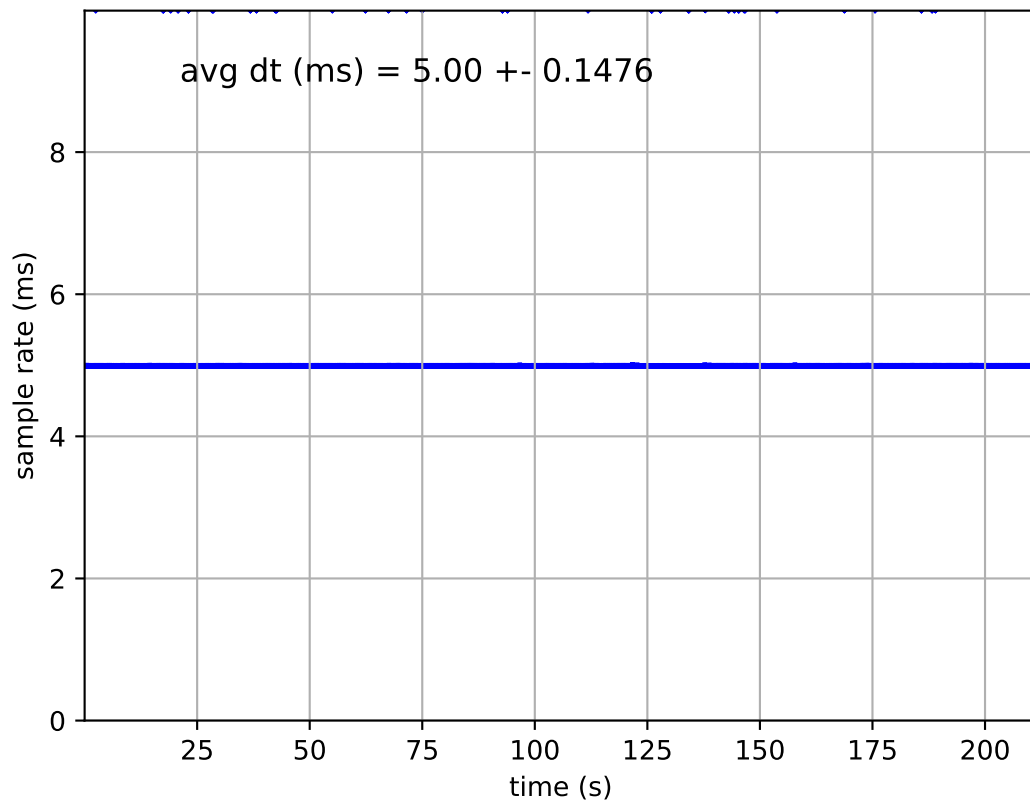
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

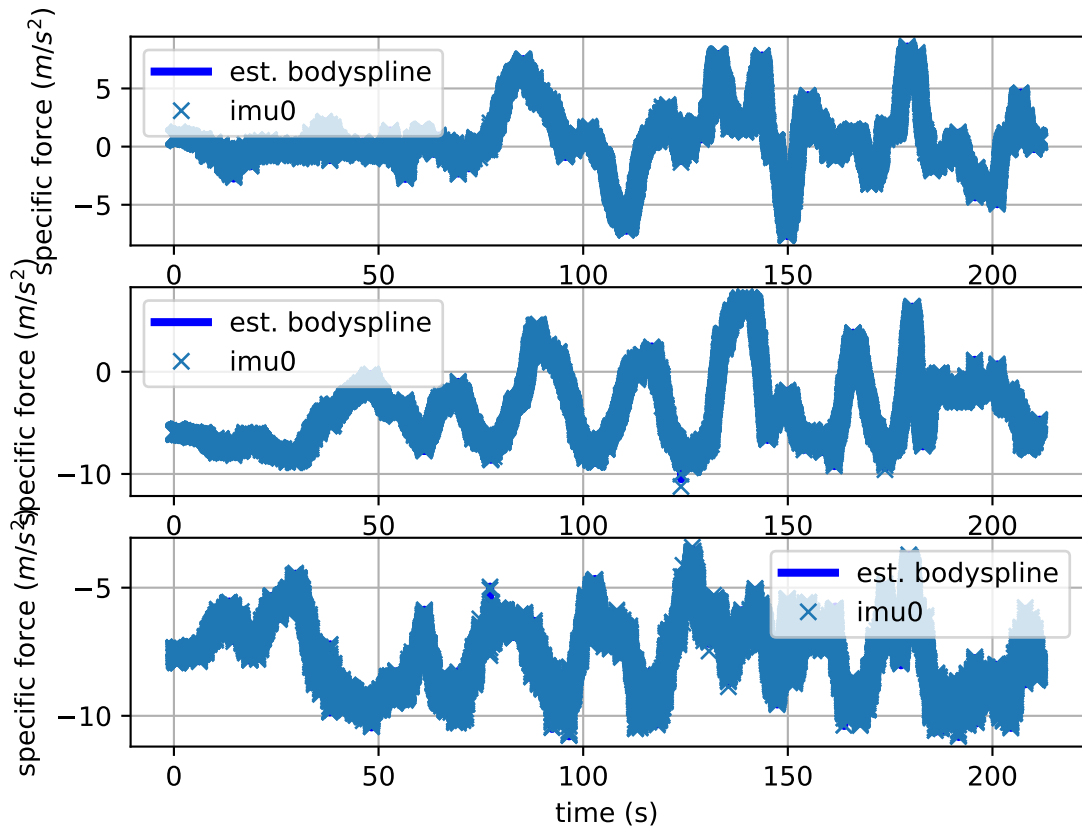
imu0: estimated poses



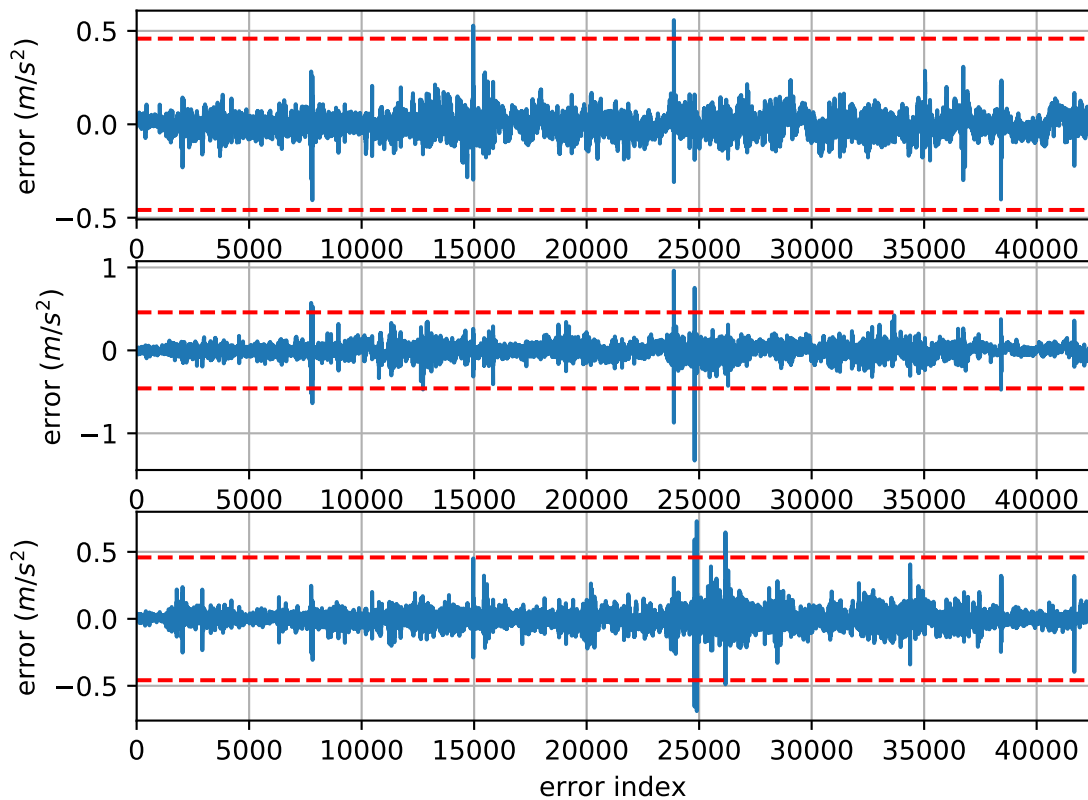
imu0: sample inertial rate



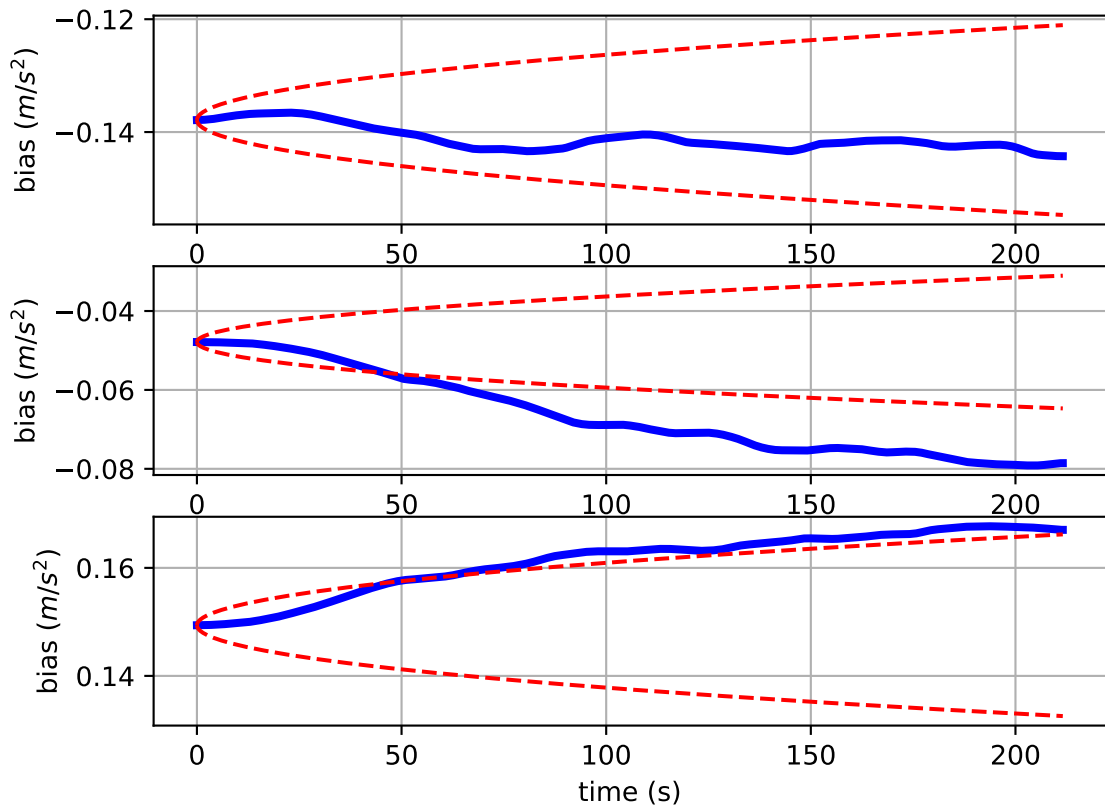
Comparison of predicted and measured specific force (imu0 frame)



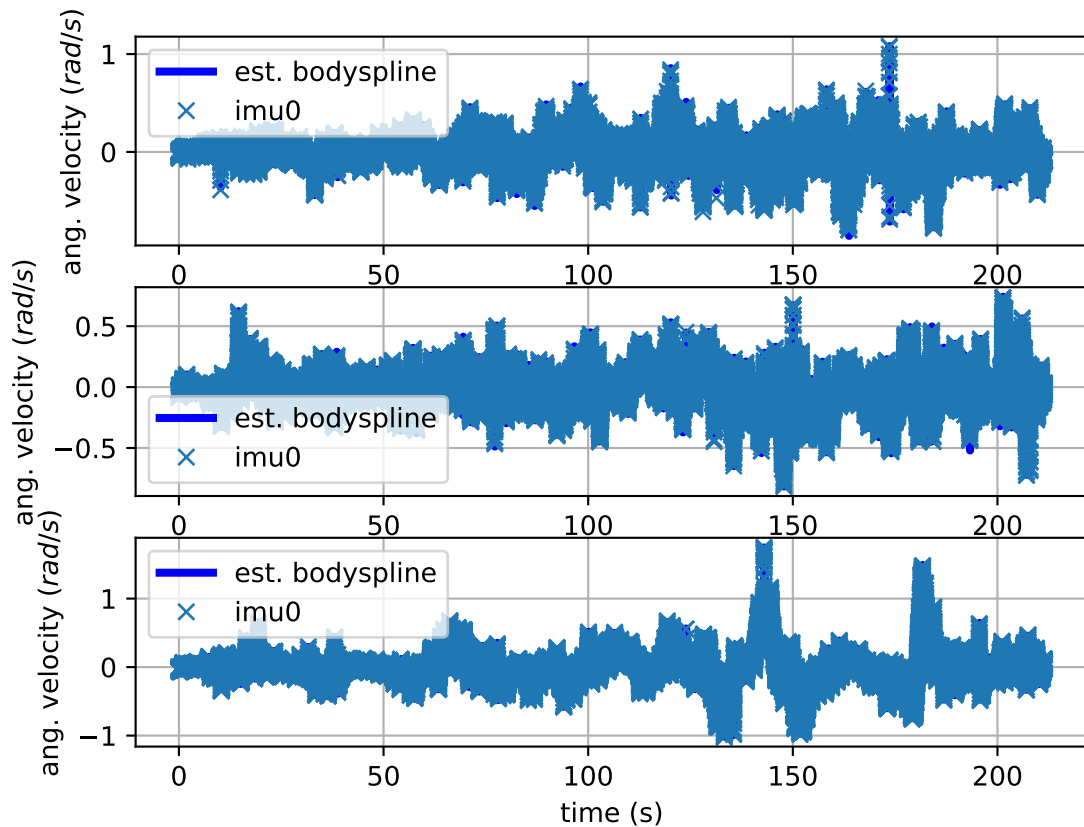
imu0: acceleration error



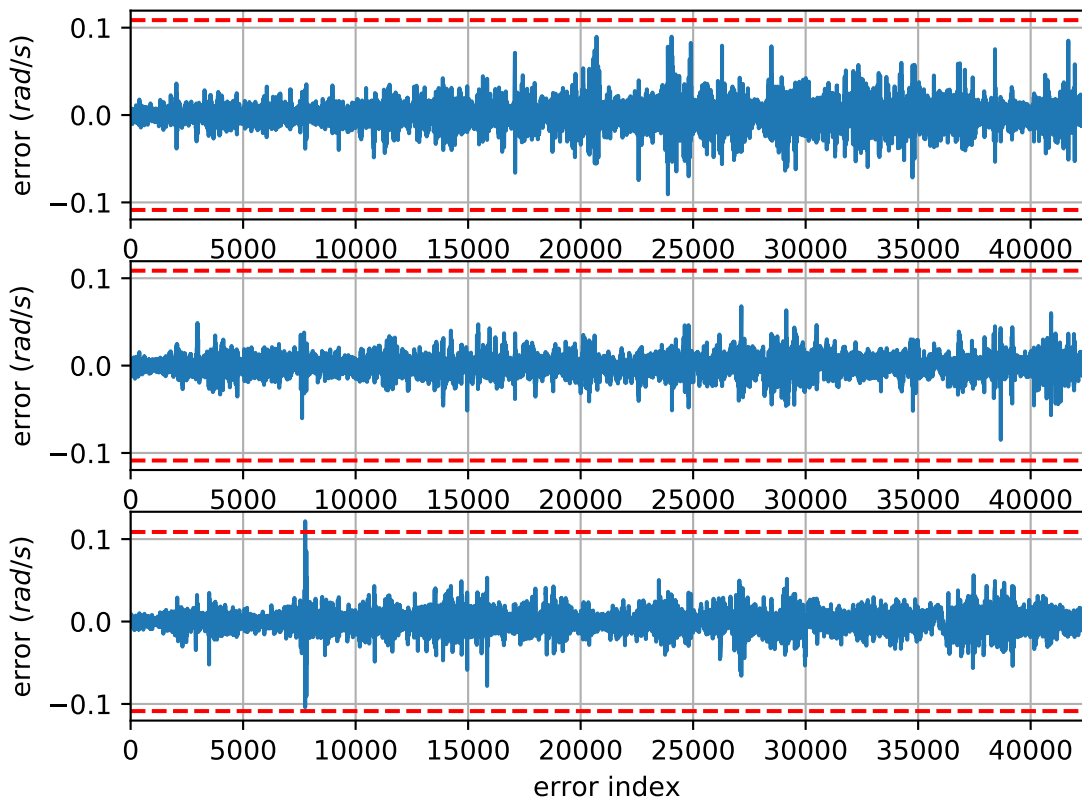
imu0: estimated accelerometer bias (imu frame)



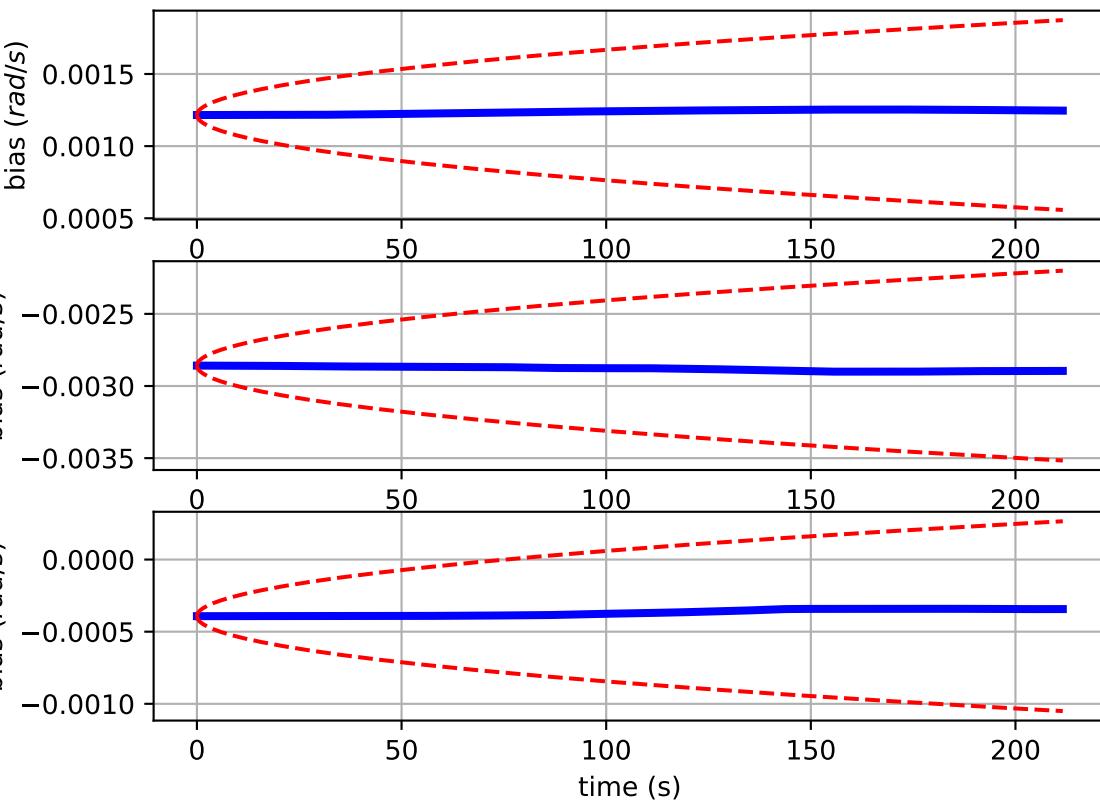
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

